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FACULTY OF COMPUTING
UTM Johor Bahru

SECP3204-01 KEJURUTERAAN PERISIAN (SOFTWARE ENGINEERING (WBL))

Semester 01, 2025/2026

PROJECT PROPOSAL

HASTA TRAVEL CAR BOOKING SYSTEM

HASTA TRAVEL & TOURS ORGANIZATION

Team Name: Hephaestus

Team Members:

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2. Ahmad Irfan Bin Azahan
3. Cheong Yi Shien
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1. Introduction

Businesses and consumers benefit from digital technology (TatvaSoft, 2022). Automating and offering online services is becoming increasingly popular in the automotive rental industry (MyRent, n.d.; TopRentApp, n.d.; WAHDAH, n.d.). Travel currently has a semi-manual process that is inefficient and error-prone. We propose a web-based car rental administration system tailored to Hasta Travel's needs (Hasta Travel, n.d.; Reepei, n.d.). In addition to increasing employee productivity and customer satisfaction, it will also improve payment verification and car return (TatvaSoft, 2022). Digital platforms allow Hasta Travel to provide its customers with higher levels of accuracy, efficiency, and reliability. Table 1 illustrates the NABC analysis for the car rental system in Hasta Travel.

Table 1: NABC Analysis – Car Rental System

Need (N)	Hasta Travel developed System Car Rental to address significant internal operational inefficiencies as well as to remain competitive in a rapidly digitizing industry. A semi-manual system dominates until Hasta Travel's current operations, with fragmented channels of communication and error-prone manual processes (Reepei, n.d.). A customer must complete an online form, then use WhatsApp to confirm bookings, submit documents, and verify payments (Reepei, n.d.). As a result of the disjointed process, response times are slower, redundant data is generated, miscommunications occur, and inconsistent data is collected (Reepei, n.d.). In the absence of a centralized platform, employees are unable to maintain accurate payment histories, monitor customer
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	interactions, or track vehicle availability in real time (Heroku, n.d.; TatvaSoft, 2022). It hinders internal productivity as well as leads to a suboptimal customer experience characterized by delays and lack of transparency (TatvaSoft, 2022). Moreover, management does not have access to consolidated data sources, which means it cannot make data-driven, revenue-driven decisions (Heroku, n.d.; TatvaSoft, 2022). In addition to internal operational challenges, market forces make this system more important (TatvaSoft, 2022). In the automobile rental industry, automation and integrated online services are clearly on the rise (MyRent, n.d.; TopRentApp, n.d.; WAHDAH, n.d.). Digital interactions with service providers must be efficient, secure, and streamlined (TatvaSoft, 2022). Automated platforms and sophisticated digital platforms are already offered by competitors (MyRent, n.d.; TopRentApp, n.d.; WAHDAH, n.d.). As a result, Hasta Travel suffers from a serious competitive disadvantage because of its outdated, semi-manual system (MyRent, n.d.; TopRentApp, n.d.; WAHDAH, n.d.). By failing to modernize, it threatens customer retention and long-term viability (TatvaSoft, 2022). Therefore, digital car rental management is more than just an operational upgrade; it is a strategic adjustment to ensure business
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	sustainability and competitive parity (Heroku, n.d.; TatvaSoft, 2022).
Approach (A)	As a solution, Hasta Travel's existing website will be enhanced with a bespoke web-based car rental administration system (Hasta Travel, n.d.). In addition to Laravel and MySQL, this system will be built using cost-effective, open-source technologies (Heroku, n.d.; TatvaSoft, 2022). A portal for uploading documents and proofs of payment, along with online booking, is among the most important features (MyRent, n.d.; TopRentApp, n.d.; WAHDAH, n.d.) state that the system uses a manual proof-upload mechanism instead of integrating a third-party payment gateway in order to manage scope and cost (Reepei, n.d.; TatvaSoft, 2022).
Benefit (B)	System Car Rental is expected to have a significant impact across the organization (Heroku, n.d.; TatvaSoft, 2022). Operational staff will benefit from automated operations, reduced errors, and increased administrative efficiency (Heroku, n.d.; TatvaSoft, 2022). There is no doubt that System Car Rental is likely to have a significant impact across the organization (Heroku, n.d.; TatvaSoft, 2022). By automating operational processes, reducing errors, and increasing administrative efficiency, operational staff will benefit (Heroku, n.d.; TatvaSoft, 2022).
Competitor (C)	In the car rental software market, there are mature commercial SaaS platforms like

	MyRent and TopRentApp (MyRent, n.d.; TopRentApp, n.d.). While System Car Rental may have some similarities to these platforms, it is not intended to compete directly with them. Its strategic advantage lies in being a custom-built, cost-effective internal tool tailored specifically to Hasta Travel's needs (Hasta Travel, n.d.). The company will be able to solve its specific problems economically efficiently without the recurring subscription fees and unnecessary complexity of commercial systems.
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2. Existing Systems

The current system of Hasta Travel is handled in a traditional way, they are using web-based and allows customers to browse vehicles and submit the booking form like in Figure 1, and it directly sends the booking details to the WhatsApp to communicate with staff for booking confirmation and payment process. The payment can be accepted through QR code, bank transfer, or cash.

Once the booking form is submitted via WhatsApp, like in Figure 2, the staff will manually check the status of availability of the vehicles, then request the customer to make deposit payment. Customers need to manually upload the receipt to WhatsApp and wait for the staff to check, compare, and validate the receipt with the bank statement.

The staff must manually prepare and submit invoices. The current system does not have an automated e-invoice generator, automated reports generator and completion order customer send through email and a save copy for the company. The current system does not have real-time vehicle status availability, where the staff needs to check and notify the customer for confirmation of the vehicle's availability through WhatsApp. The system lacks feature like automatic fees calculation that are based on rental duration and dynamic pricing that varies according to promotions, weekdays, weekends or holidays. As shown in Figure 3, compared to Hasta Travel, the WAHDAH car rental system (WAHDAH, n.d.) is much more automated.

As shown in Table 2, the comparison of features between Hasta Travel system (Hasta Travel, n.d.) and WAHDAH system (WAHDAH, n.d.) highlights what features to be implemented in the proposed system.

Figure 1: Booking form of Hasta Travel (Hasta Travel, n.d.)

Car Rental Booking Form

* Car Model
This field is required
If car not in option, please mention in remarks section.

* Name
This field is required
Please fill in your full Name

* Enter your email
This field is required
We send all our best offers and discounts to our customers via email

* Matric/Staff ID
This field is required
We only provide rental to UTM staffs & students only.
If Alumni, please state in the remarks section below.

* IC / Passport
This field is required
Using this format: 057438927466 or A4567890

Enter your phone number
+60

Figure 2: Booking form submitted through WhatsApp (Hasta Travel, n.d.)

Hasta Booking
Response #10370

Car Rental Booking Form

Name : MUHAMMAD HAFIZ BIN REEPEI
Enter your email : muhammadhafiz.r@graduate.utm.my
Matric/Staff ID : A24CS0133
IC / Passport : 050207050167
Enter your phone number : 60134976641
Address : MA1 WING-A, KTDI
Faculty : FC
Car Model : Others
Pickup Date & Time : 02-06-2025, 10:00 pm
Pickup Location : Student Mall, UTM
Return Date & Time : 03-06-2025, 10:00 pm
Return Location : Student Mall, UTM
Rental Duration : 1 day
Rental (RM) : 200
Destination : Gunung Pulai, JB
Remarks : Alza

Thank You for contacting Hasta Travel & Tours Sdn Bhd 🙏

Please fill in this form for us to check your car availability:
<https://whatsform.com/8bkqoa>
or visit our website
<https://hastatravel.com>

31/05/2025

Figure 3: Automated fee calculation from WAHDAH system (WAHDAH, n.d.)

The screenshot displays the WAHDAH payment interface. On the left, under 'DRIVER INFO', the driver's name is MUHAMMAD HAFIZ BIN REEPEI, with an IC number 050207050167, HP number +60134976641, license number 237482397, and age 27. The 'PAYMENT METHOD' section shows options for Debit/Credit Card (highlighted), Online Banking, Paypal, and WaCash. Logos for VISA, Mastercard, and DISCOVER are visible. A 'Proceed To Payment' button is at the bottom. On the right, the 'Summary of Charges' table lists: Rental (125.00), Pick Up Fee (34.44), Return Fee (34.44), and Add-ons (0.00). It also shows a Membership Discount of -0.00, a Total Amount of 193.88, Rounding of 0.00, and a Refundable Deposit of 300.00. The Grand Total (MYR) is 493.88. A note mentions a 10-day deposit withdrawal period, and a link for promo codes is provided.

Table 2: Comparison of existing systems

Features	Hasta Travel	WAHDAH	Proposed System
User registration	No	Yes	Yes
Dynamic Pricing	No	No	Yes
Real-time vehicle availability statuses	No	Yes	Yes
Booking and payment through system	No	Yes	Yes
Automated fees calculation	No	Yes	Yes
Central Database	No	Yes	Yes
Membership Discount	No	Yes	Yes
Loyalty Card	No	No	Yes

3. Proposed System

The system is a part of a larger system that's designed to overcome the limitations of the existing system by providing a more efficient alternative. It aims to make system operations easier to use and enhance user experience.

One of the users involved is the customer who can register and log in to the system to book a car where they can upload documents, view available vehicles in real time, and quicker payments. Also, their transactions are rentals recorded in the central database for future references. The customers are also assisted by the Auto Calculation Price Systems to help them know the booking price based on their rental time. When a customer pays for a car rental, the system automatically logs the amount, date, and reference number into the accounting database that will automatically record payment transactions, generate digital invoices, and update financial records in real time. The customer can also claim rewards from the loyalty system for the rentals they booked.

Apart from customers, the car owner is also a user of the system. They can use the system to register their car and update the car details such as model, condition, and availability. With this, the system helps them monitor their cars easily.

Lastly, the admins are the most frequent users using the system where their job is checking customer details, approving or rejecting bookings, and updating car statuses. They also need to manage deposit records, assign pickup or delivery tasks, and make sure all transactions are properly stored in the database. They can also access reports that include sales, bookings, and customer feedback that can help them analyze performance to improve the sales and customer experience.

4. Software Process Model

The software process model that we use to develop the Hasta Car Booking System is the agile model. The flexibility and iteration approach to software development of agile models lead an easy adaptation towards changing requirements. The agile methodology model allows a close collaboration between development team and client which thus helps to collect frequent feedback in order to ensure a satisfying system that fits client requirements is produced.

Unlike traditional models, agile involves multiple sprints which each containing planning, analysis, design, implementation and testing activities until the system meets user's needs. Each sprint produces a functional increment of the system, reviewed and refined based on client feedback. The team conduct weekly stand-ups meetings to ensure every progress is smoothly carried out across modules. All activities should be done in a limited duration for every phase with certain deliverables and every member is assigned with tasks and roles as shown in Table 2. At the same time, every member will handle a sprint module and list of tasks in-charge are shown in Table 3. Table 4 shows all activities done for each module for each member during implementation phase.

Every member helps with analysis, design, and testing for all modules. During implementation stage, each member will handle different modules. All members should join a weekly stand-up to upgrade the progress of different modules.

Table 3: Agile SDLC for Hasta Car Rental System

Phases	Activities	Duration	Person in-charge	Deliverable
Planning	• Conduct initial meetings with clients to understand the project goals • Identify key features • Prepare user story • Define sprint, role and project schedule • Estimate time and resources	Week 1 – Week 3	All members	• Problem definition and objective summarize • Project proposal • User story • Sprint plan

Analysis	• Gather detail functional and non-functional requirements • Analyze business rules and data needs • Create data flow diagram • Define system use case • Refine and prioritize backlog with client feedback	Week 4 – Week 6	All members	• Requirement specification document • Data Flow Diagram • Use case diagram
Design	• Design system architecture and database • Create UI wireframes for user interface • Define relationship between modules • Define navigation flow and integration points • Review and update design based on feedback	Week 7 – Week 10	All members	• System design document • ER diagram • UI mockups • Architecture layout
Implementation	• Develop system modules in multiple sprints: - Module 1: Vehicle Management and Car Booking Module -Module 2: Customer Module & Booking	Week 11 – Week 17	All members	• Functional system prototype • Source code • Database schema

	Design -Module 3: Admin & Reporting Modules and Booking cancellation -Module 4: Payment & Billing Module • Develop front-end interface • Implement back-end logic • Integrate database and API connection • Conduct code review and sprint demonstration after each sprint • Integrate and test module			
Testing	• Conduct unit, integration and system testing • Perform user acceptance testing • Fix bugs and optimize performance • Prepare final presentation documentation	Week 18 – Week 21	All members	• Test report • Bug log • Final working system • User manual

Table 4: Module distribution and member task distribution

Member	Task	Deliverables
Member 1 – Muhammad Hafiz Bin Reepei (Project Leader) Module 1: Vehicle Management and Car Booking Module	• Create, add, update, delete car detail functions (model, plate number, price, status) • Real-time car inventory updates of car availability • Design and implement list of car available interface • Develop booking interface (Select car, date, duration) • Design and implement the summarize for all reservations on certain date	• Car management interface • Database table for cars • Database of car • Database of owner • Database of car brand • Working real-time available car update • Working car list interface • Working Booking page • Database for bookings • Summarize of reservation based on date • Test report for customer module
Member 2 – Ahmad Irfan Bin Azahan Module 2: Customer Module & Booking Design	• Design and implement user registration and login form • Design and implement customer documentation upload page and personal detail form • Design and implement the capture upload recognize update technology • Design and implement profile page • Test user and booking workflow	• Working user registration and login form - Customer documentation upload page - Customer personal detail form • Database of customer detail - Name, identity card number, matrices number, email, faculty, etc. • Working and editable profile management page

	• Design and implement booking form	• Database table for customer details • Test report for customer module • Working booking form
Member 3 – Cheong Yi Shien Module 3: Admin & Reporting Modules and Booking cancellation	• Develop admin dashboard • Create rental report pages • Design and implement analysis and charts generation feasibility • Handle user roles and access control • Design and implement an automated email send after successful booking • Design and implement an automated notification for new reservations and cancellations. • Design and implement booking cancellation logic • Integrate all modules and perform system testing	• Admin dashboard • Rental report page - Monthly Rental - Year Rental - Faculty Rental - Car Brand Rental • Charts • Automated email • Automated notification • Booking cancellation • Final system testing and documentation
Member 4 – Ling Yu Qian Module 4: Payment & Billing Module	• Calculate rental cost • Design charges algorithms and logic • Design payment interface • Design invoices and receipts • Design and implement dynamic pricing logic	• Working cost calculation feasibility • Algorithms and logic for rental fees, late charges, etc. • Payment interface • Invoices and receipts designs • Working dynamic pricing feasibility

	• Integrate payment confirmation into booking flow • Perform test on calculation and transaction • Auto submit the e-invoice to Inland Revenue Board of Malaysia (IRBM).	• Working receipt upload feasibility • Database of invoices • Database of payments status • E-invoice to IRBM • Payment validation test report
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Table 5: Week Project plan

Phase	Week	Member 1	Member 2	Member 3	Member 4
Planning	1	Discuss project scope and objective	Gather initial requirements	Contribute ideas for report	Suggest payment features
	2	Draft user stories and sprint plan	Define user role and customer needs	Define admin and reporting requirements	Define payment and invoice requirements
	3	Finalize sprint schedule and task assignment	Help prepare project proposal	Prepare user story documentation	Review project proposal and resource allocation
Analysis	4	Identify functional requirements (booking & card management)	Gather customer-related data requirements	Gather admin/report requirements	Analyze billing and transaction requirements
	5	Create data flow and use case for booking	Create data flow and use case for customer	Define admin access use case	Define payment process use case
	6	Refine backlog with team and client	Refine backlog with team and client	Refine backlog with team and client	Refine backlog with team and client
Design	7	Design ER diagram for car and booking tables	Design ER diagram for customer table	Design ER diagram for reports and admin tables	Design ER diagram for payment and invoice tables

	8	Design car management UI wireframes	Design registration and profile UI	Design admin dashboard UI	Design payment interface UI
	9	Finalize booking page layout and logic	Finalize customer data input layout	Design report and chart layout	Design billing and invoice page layout
	10	Conduct sprint design review	Conduct sprint design review	Conduct sprint design review	Conduct sprint design review
Implementation	11	Start coding car management (add, edit, delete)	Start coding registration & login	Prepare admin dashboard structure	Develop payment interface skeleton
	12	Develop booking module (date, duration, summary)	Develop profile management & document upload	Implement basic report table	Create payment calculation logic
	13	Connect booking system to database	Connect registration system to database	Connect report table to database	Connect payment interface to database
	14	Test car and booking features	Test registration and profile update	Develop chart and analytics function	Develop invoice design and e-invoice logic
	15	Add validation to booking form	Add validation to customer data	Implement automated email for booking success	Implement payment validation

	16	Conduct sprint demo (booking module)	Conduct sprint demo (customer module)	Conduct sprint demo (report module)	Conduct sprint demo (payment module)
	17	Integrate booking with admin module	Integrate customer with booking module	Integrate admin and booking cancellation	Integrate payment confirmation with booking
	18	Perform unit testing on booking	Perform unit testing on registration	Perform testing on report and admin dashboard	Perform testing on payment and invoice
Testing	19	Fix bugs in booking and car module	Fix bugs in customer module	Fix issues in admin report generation	Fix bugs in payment module
	20	Conduct full system integration testing	Conduct full system integration testing	Conduct full system integration testing	Conduct full system integration testing
	21	Final sprint review and presentation	Final sprint review and presentation	Final sprint review and presentation	Final sprint review and presentation

5. Project Schedule

SDLC phases involved are assigned with tasks for each member. The tasks and duration for each member are converted into the Gantt chart. Figure 4 shows the project Gantt chart based on group and overall planning. Figure 5 and Figure 6 show the project Gantt chart on detailed tasks assigned to every member on certain week.

Figure 4: Project Gantt chart based on Group

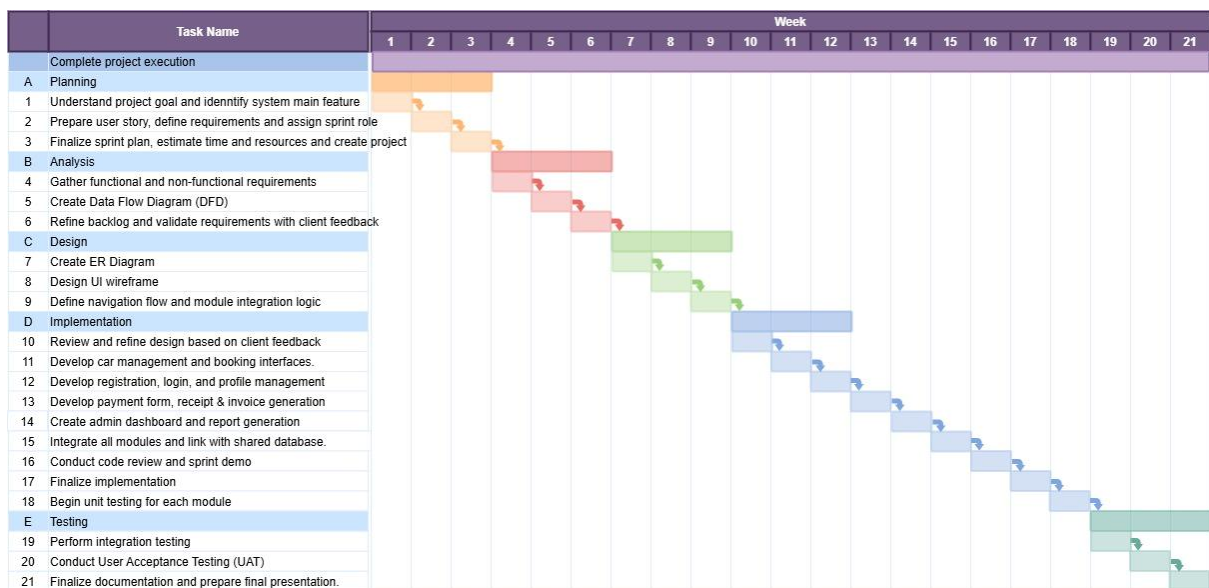


Figure 5: Project Gantt chart based on Member (Planning, Analysis, Design phase)

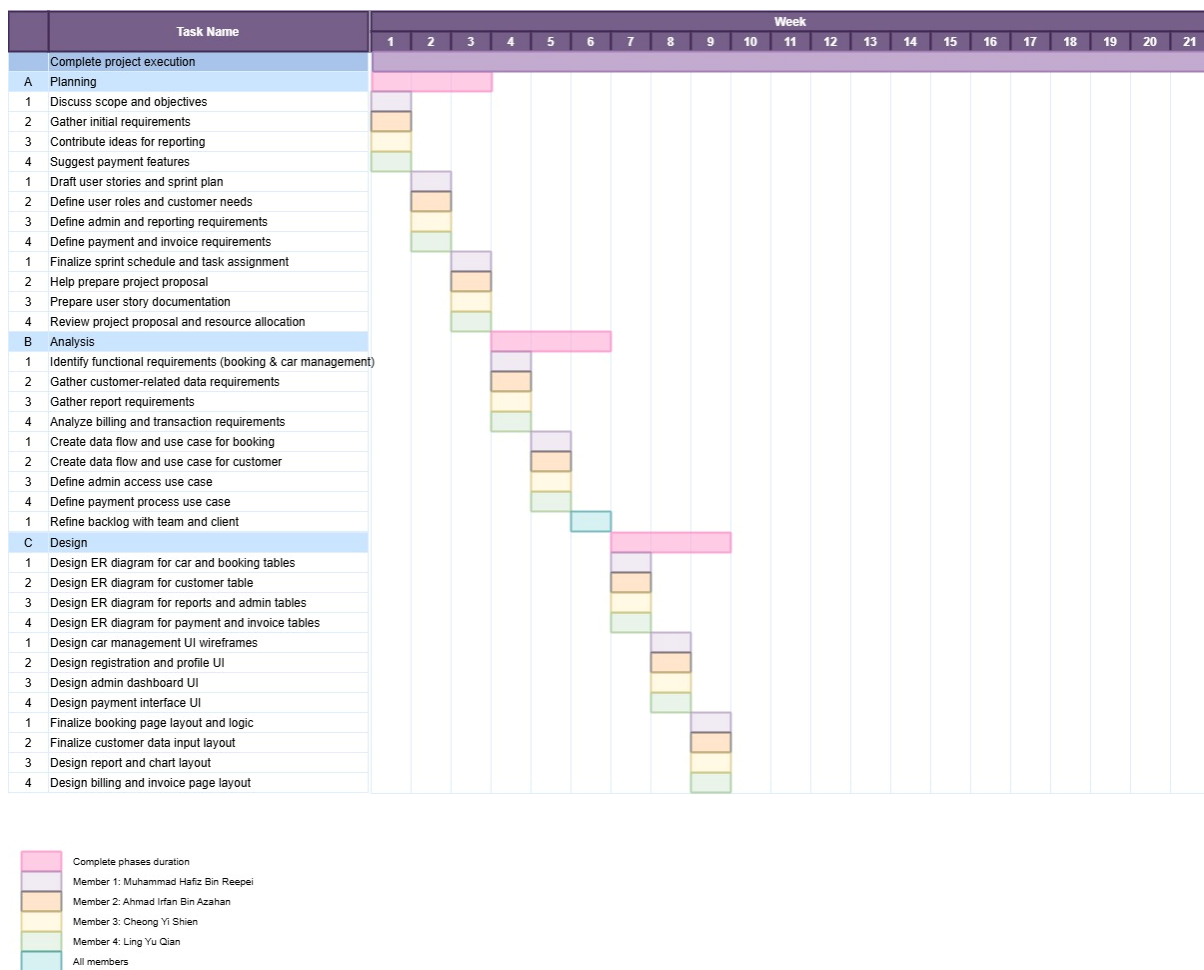
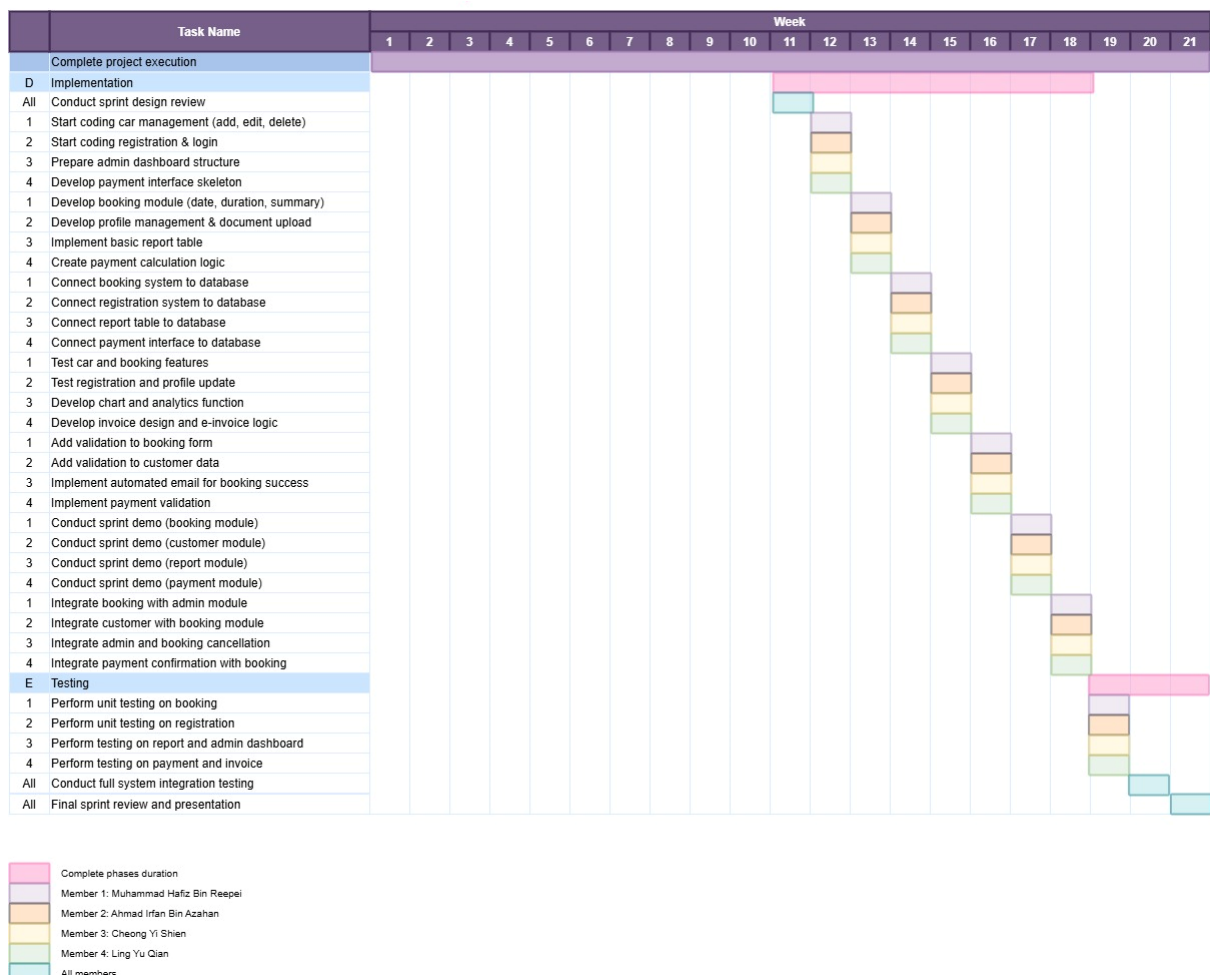


Figure 6: Project Gantt chart based on Member (Implementation, Testing phase)



6. References

- [1] Hasta Travel. (n.d.). *Car rental*. Retrieved from <https://hastatravel.com/car-rental>
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- [6] TopRentApp. (n.d.). *Car rental software*. Retrieved from <https://toprentapp.com/>
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Appendices

Update: Turnitin Update Oct 14, 2025

What's new?

- [NEW] Turnitin AI writing model got updated Oct 14, 2025
- Ai detection now works for English, Spanish & Japanese language
- Ai checking Criteria Updated [?]
- Ai detection score between 1%-19% will be marked with *% without any highlights [?]
- Get highlights feature- Get AI highlights even when you get *% (**note: this feature uses 1 extra slot**)

 Reports

☐ Exclude biblio

Unique ID	Submitted file	AI	Similarity	Status
2527963	2024.SECJ2203_SE_Assignment_1...pdf	*%	8%	completed

Figure 4: Screenshot of checking generative AI using Turnitin

Yes — there are many online examples of project proposals for systems, and they can be very useful for you to model your own. Here are a few good ones, along with what to look for and how to use them for your needs.

✓ Some good example sources

- This article “How to Write a Project Proposal (Examples & Template Included)” offers a breakdown of what a project proposal should contain. [ProjectManager](#)
- “Top 10 System Proposal Examples with Samples and Templates” gives system-specific proposal examples (which is very relevant if your project is a software/system project). [slideteam.net](#)
- A full example: *Project Proposal: An Enterprise Information System for retail companies’ internal inventory management*. This is a concrete system proposal (IT/system based). [elc.polyu.edu.hk](#)
- A “Project Proposal System Analysis and Design” example for IT students. [Scribd](#)

Figure 5: Screenshot of the use of Generative AI Tool using ChatGPT for Project Proposal Example

[URL: https://chatgpt.com/s/t_69075e46ec6881919728f05fbeb141c7]

Cool — here are **six relevant car-rental competitors** around Johor/Southern Malaysia you can dig into for your market research for Hasta Travel Car Rental. I'll include what they do well so you can pick up ideas.

1. WAHDAH

★ 4.8 • Car rental agency



What they do well:

- They highlight **well-maintained cars under 5 years old**. [wahdah.my +1](#)
- They provide 24/7 roadside assistance. [wahdah.my +1](#)
- Transparent pricing: “What you see is what you pay”. [wahdah.my](#)

Why useful for you:

- For Hasta Travel Car Rental, emphasising fleet condition + transparent pricing can be a differentiator.
- The after-sales service (roadside assistance) is a value add you might incorporate.

Figure 6: Screenshot of the use of Generative AI Tool using ChatGPT for six relevant car-rental competitors

[URL: <https://chatgpt.com/share/69076390-3260-800b-9d92-b73773882808>]

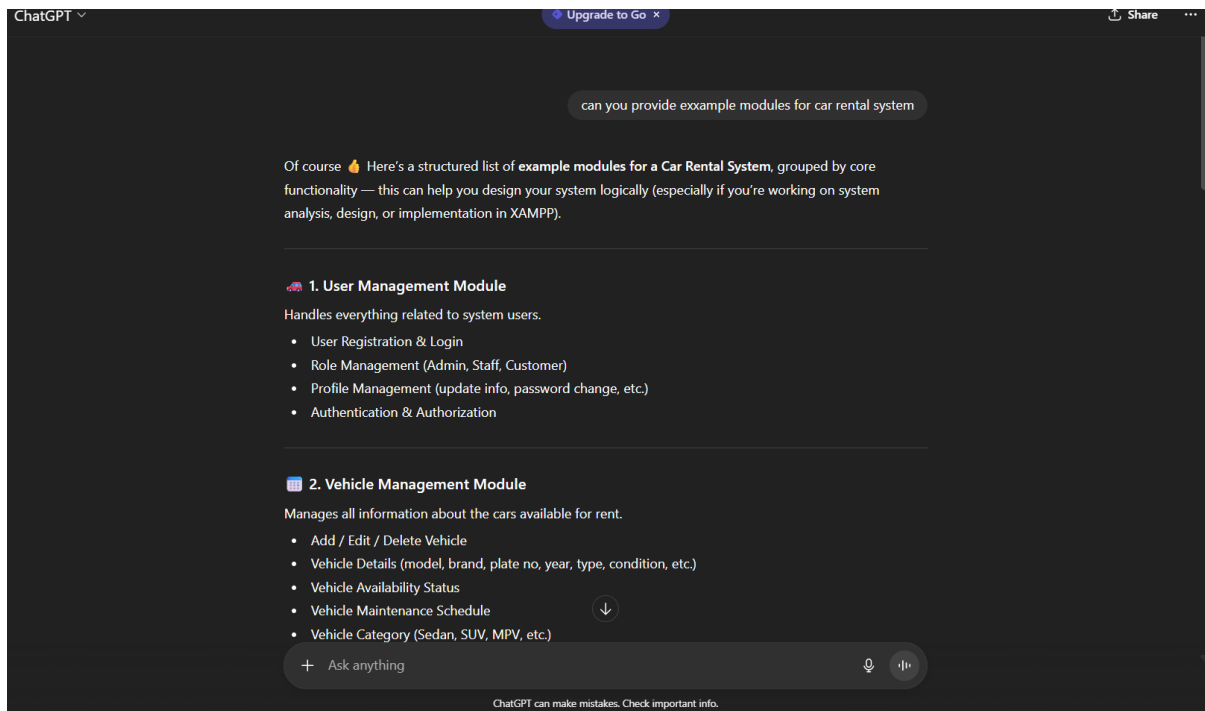


Figure 7 : Screenshot from ChatGPT for the module's car system.

[URL: <https://chatgpt.com/share/69075ece-2024-8013-b26e-6766e39c9288>]